

TEST YOUR UNDERSTANDING

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Levels 1-2: Remembering/Understanding

- The *largest* unit within which gene flow can readily occur is a
(A) population. (C) genus.
(B) species. (D) hybrid.
- Males of different species of the fruit fly *Drosophila* that live in the same parts of the Hawaiian Islands have different elaborate courtship rituals. These rituals involve fighting other males and making stylized movements that attract females. What type of reproductive isolation does this represent?
(A) habitat isolation (C) behavioral isolation
(B) temporal isolation (D) gametic isolation
- According to the punctuated equilibria model,
(A) given enough time, most existing species will branch gradually into new species.
(B) most new species accumulate their unique features relatively rapidly as they come into existence, then change little for the rest of their duration as a species.
(C) most evolution occurs in sympatric populations.
(D) speciation is usually due to a single mutation.

Levels 3-4: Applying/Analyzing

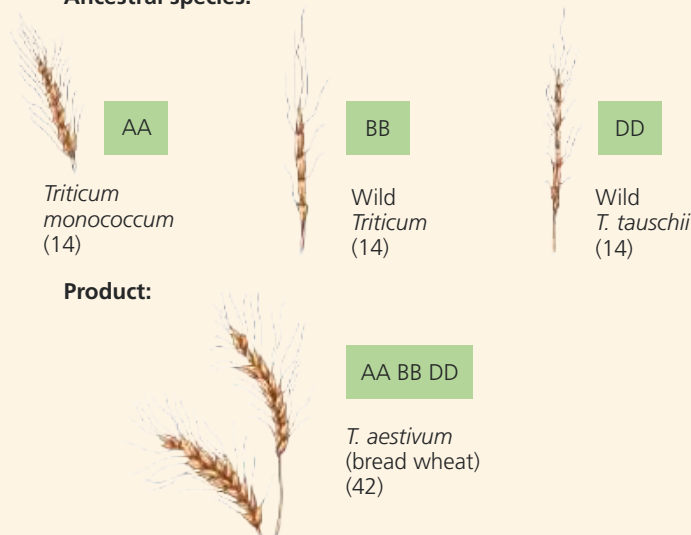
- Bird guides once listed the myrtle warbler and Audubon's warbler as distinct species. Recently, these birds have been classified as eastern and western forms of a single species, the yellow-rumped warbler. Which of the following pieces of evidence, if true, would be cause for this reclassification?
(A) The two forms interbreed often in nature, and their offspring survive and reproduce well.
(B) The two forms live in similar habitats and have similar food requirements.
(C) The two forms have many genes in common.
(D) The two forms are very similar in appearance.
- Which of the following factors would be the most likely to contribute to allopatric speciation?
(A) The separated population is large, and genetic drift occurs.
(B) Selection pressures in the isolated population are similar to those in the ancestral population.
(C) Gene flow between the two populations is extensive.
(D) Different mutations begin to distinguish the gene pools of the separated populations.
- Plant species A has a diploid chromosome number of 12. Plant species B has a diploid number of 16. A new species, C, arises as an allopolyploid from A and B. The diploid number for species C would probably be
(A) 14. (C) 28.
(B) 16. (D) 56.

Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** Explain the biological basis for assigning all human populations to a single species. Can you think of a scenario by which a second human species could originate in the future?
- SCIENTIFIC INQUIRY • DRAW IT** In this chapter, you read that bread wheat (*Triticum aestivum*) is an allohexaploid, containing two sets of chromosomes from each of three different parent species. Genetic analysis suggests that each of the three species pictured following this question

contributed chromosome sets to *T. aestivum*. (The capital letters represent sets of chromosomes, each of which can be traced to a particular species, *not* individual genes. The numbers in parentheses are the species' chromosome numbers.) Evidence also indicates that the first polyploidy event began with the spontaneous hybridization of the early cultivated wheat species *T. monococcum* and a wild *Triticum* grass species. Draw a diagram of one possible chain of events that could have produced the allohexaploid *T. aestivum*.

Ancestral species:



- WRITE ABOUT A THEME: INFORMATION** In sexually reproducing species, each individual inherits DNA from both parent organisms. In a short essay (100–150 words), apply this idea to what occurs when organisms of two species that have homologous chromosomes mate and produce (F_1) hybrid offspring. What percentage of the DNA in the F_1 hybrids' chromosomes comes from each parent species? As the hybrids mate and produce F_2 and later-generation hybrid offspring, describe how recombination and natural selection may affect whether the DNA in hybrid chromosomes is derived from one parent species or the other.

10. SYNTHESIZE YOUR KNOWLEDGE



Suppose that females of one population of strawberry poison dart frogs (*Dendrobates pumilio*) prefer to mate with males that are orange-red in color. In a different population, females prefer males with yellow skin. Explain how such differences could arise and how they could affect the evolution of reproductive isolation in allopatric versus sympatric populations.

For selected answers, see Appendix A.